地政總署
Lands Department

AECOM

Simplified Data Specifications for

TRANSPORTATION FSDT

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Foreword

To promote better sharing and accessibility of spatial data among the community of stakeholders in Hong Kong Special Administrative Region ("HKSAR"), the Government acknowledged its commitment to establishing Common Spatial Data Infrastructure ("CSDI"). The main objective is to provide Bureaux/Departments ("B/Ds") and public and private organisations with an information infrastructure to share spatial data, support various smart city applications, and support Smart City development.

In March 2017, the Development Bureau ("DEVB") commissioned a consultancy study on the overall strategy for implementing the CSDI. With its implementation, the CSDI would benefit the Government and society in the long run, with convenient and easy access to high-quality and up-to-date spatial information. The CSDI will help to:

- Facilitate and support the development of a dynamic and flexible geospatial environment that will provide standardised interfaces for accessing and disseminating common spatial information;
- Facilitate information sharing across different B/Ds and eventually other stakeholders;
- Encourage the adoption of common standards to ensure quality, compatibility and interoperability amongst different systems; and
- Make available a wide range of spatial data and analyses to enable potential applications.

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1. Introduction

1.1. General Overview

The Simplified Data Specifications for Transportation ("the/this document") provide information about the datasets in the Transportation Framework Spatial Data Theme ("FSDT").

1.2. Glossary and Definitions

This section provides a glossary of terms and definitions, listed alphabetically, as applicable to this document.

- **Feature**: Abstraction of real-world phenomena. A feature may occur as a type or an instance.
- **Feature Attribute**: Characteristic of a feature. A feature attribute has a name, a data type, and a value domain associated to it.
- **Line**: Topological 1-dimensional geometric primitive, representing a linear feature.
- **Point**: Topological 0-dimensional geometric primitive, representing a position.
- **Polygon**: Topological 2-dimensional geometric primitive, representing an area / a planar surface.

1.3. Acronyms and Abbreviations

This section lists the expanded description of all acronyms (words formed from the initial letters of a phrase) and abbreviations (the shortened form of a word or phrase) used in this document in alphabetical order.

Acronym/Abbreviation	Description
B/D	Bureau/Department
CSDI	Common Spatial Data Infrastructure
DEVB	Development Bureau
FSDT	Framework Spatial Data Theme
GIS	Geographic Information System
HKSAR	Hong Kong Special Administrative Region
ISO (or ISO/TS)	International Organization for Standardization (or ISO/Technical Specifications)
iB1000	Digital Topographic Map iB1000
iRG1000	Road Centre Line
LandsD	Lands Department
LIS	Land Information System
SI	International System of Units
TD	Transport Department
the/this document	Simplified Data Specifications for Transportation

2. Framework Spatial Data Definitions

2.1. Overview

Transportation data should comprise an integrated and seamless transport network with related features. Typical geographic topics in a transportation infrastructure Geographic Information System ("GIS") dataset include roads, rails, cables, waters and air transport networks. To support the requirements for intelligent transport systems as well as smart mobility applications and services, it is important that transportation features form geometric networks when appropriate, and connection links between different networks are established if possible. The transport network should also support referencing transport flows and directions to enable navigation services.

Transportation information can also be used in various spatial analysis and for general reference. For example, if properly modelled the street network can be used to route delivery trucks, school buses, public transport vehicles, emergency medical or police response, and other routing applications. It can also be used for network and proximity analysis, by correlating the street network with additional information, such as the population adjacent to the streets to be served by public transportation, and other factors.

2.2. Scope & Definitions

This document covers the following sub-themes:

- **Road Transportation:** Road Transportation includes the road network, such as traffic directions, turning restrictions at road junctions and other related road traffic data that may support navigation.
- **Rail Transportation:** Rail Transportation includes the physical and non-physical components representing the rail mode of travel that allow the movement of goods and people between locations, supporting infrastructure necessary for rail operations and maintenance and related data to support navigation purposes.
- **Water Transportation:** Water Transportation includes water-based routes for boats, shipping channels, ferries, container ship berths, and other forms of water transportation, including cruise terminals, ferry terminals, stations, harbours, marinas, boat launches, etc. Data included should support users' decisions for navigation.
- **Air Transportation:** Air Transportation includes air travel facilities such as airports and helipads, flight paths and primary air routes that supports navigation. In contrast, the operation details inside airports will not be covered.

3. Framework Spatial Data Content & Structure

3.1. Datasets under Sub-themes

There are four sub-themes within the Transportation FSDT, namely Road Transportation, Rail Transportation, Water Transportation and Air Transportation.

Road Transportation sub-theme covers the following dataset(s):

- Road Centreline: The Road Centreline dataset is depicted based on the Land Information System ("LIS") 1:1000 Topographic dataset and is available as the Street Centre Lines feature class in the iB1000 dataset and Road Centreline in iRG1000 dataset. The usage of the LandsD's Road Centreline is primarily focused on approximate location query and map annotation labelling.
- Road Network: The Road Network dataset includes traffic directions, turning restrictions at road junctions, stopping restrictions and other road traffic data for supporting the development of intelligent transport system, fleet management system and car navigation etc.

Rail Transportation, Water Transportation and Air Transportation sub-themes do not contain any dataset(s).

3.2. Physical Data Model

This section describes the physical data model for each sub-theme within the Transportation FSDT. Due to the substantial number of fields, the diagrams have been included as Appendix I.

3.3. Entity Description

This section provides a detailed description of each physical entity illustrated in the physical data model in Appendix I.

3.3.1. Road Transportation – Road Centreline

3.3.1.1. RoadCentreLine

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
STREETCENTRELINEID	Long, 10	Street Centreline identifier	N
STREETCODE	Long, 5	Street identifier	N
STREETTYPE	String, 255	Street Type	N
ENGLISHSTREETNAME	String, 100	Street name (English)	Y
CHINESESTREETNAME	String, 100	Street name (Traditional Chinese)	Y
LASTUPDATEDATE	Date	Last updated date	N

3.3.2. Road Transportation – Road Network

3.3.2.1. VEHICLE_RESTRICTION

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
ROAD_ROUTE_ID	Long, 6	This is the field that stores the Road Route identifier of the Road Route (ROUTE_ID) being referenced.	N
MAX_AXLE_WEIGHT	String, 10	The maximum weight of the axle of the vehicle that the road can support.	Y
VR_ID	Long, 6	A unique identifier of storing the Vehicle Restriction feature.	N
MAX_LENGTH	String, 10	The maximum length of the vehicle that the road segment can support.	Y
MAX_WEIGHT	String, 10	The maximum weight of the vehicle that the road can support.	Y
MAX_HEIGHT	String, 10	The maximum height of the vehicle that the road can support.	Y
MAX_WIDTH	String, 10	The maximum width of the vehicle that the road can support.	Y
REMARKS	String, 255	This field stores additional information on Vehicle Restriction. This field can be used for displaying labels.	Y
CRE_DATE	Date	To store the date that the record was created.	N
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N
BOUND	Long, 1	This field is used to indicate the location of the feature along the centerline. If it is located on the left hand side of the centerline with reference to digitized direction, the value is 1. If it is located on the right hand side of the centerline with reference to digitized direction, the value is -1. If it is located on both sides of the centerline, with reference to digitized direction, the value is 0.	N

3.3.2.2. TRAFFIC_FEATURES

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
FEATURE_TYPE	Short, 1	There are four types of traffic features. 1 = Zebra Crossing 2 = Yellow Box 3 = Toll 4 = Cul-de-sac	N
FEATURE_ID	Long, 6	Unique identifier for traffic feature. The identifier will be used to link with external tables for other systems.	N
CRE_DATE	Date	To store the date that the record was created.	N
RD_ID_1	Long, 6	The identifier of 1st intersected Road Centreline	Y
RD_ID_2	Long, 6	The identifier of 2nd intersected Road Centreline	Y
RD_ID_3	Long, 6	The identifier of 3rd intersected Road Centreline	Y
RD_ID_4	Long, 6	The identifier of 4th intersected Road Centreline	Y
RD_ID_5	Long, 6	The identifier of 5th intersected Road Centreline	Y
RD_ID_6	Long, 6	The identifier of 6th intersected Road Centreline	Y
RD_ID_7	Long, 6	The identifier of 7th intersected Road Centreline	Y
RD_ID_8	Long, 6	The identifier of 8th intersected Road Centreline	Y
RD_ID_9	Long, 6	The identifier of 9th intersected Road Centreline	Y
REMARKS	String, 255	This field stores additional information on Traffic Features information.	Y
TUN_BRIDGE_ID	Long	To store the Feature identifier of TUN_BRIDGE_FARE table	Y
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP	N

3.3.2.3. SPEED_LIMIT

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
SPEED_LIMIT_ID	Long, 6	An unique identifier of storing the Speed Limit feature	N
ROAD_ROUTE_ID	Long, 6	This is the field that stores the Road Route identifier of the Road Route (ROUTE_ID) being referenced.	N
SPEED_LIMIT	String, 255	This is the field that stores speed limit for the route segment (in km/hr).	N
REMARKS	String, 255	This field stores additional information on Speed Limit. This field can be used for displaying labels.	Y
CRE_DATE	Date	To store the date that the record was created.	N
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N
BOUND	Long, 1	This field is used to indicate the location of the feature along the centerline. If it is located on the left hand side of the centerline with reference to digitized direction, the value is 1. If it is located on the right hand side of the centerline with reference to digitized direction, the value is -1. If it is located on both sides of the centerline, with reference to digitized direction, the value is 0.	N

3.3.2.4. RUN_IN_OUT

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
RUN_IN_OUT_ID	Long, 6	Unique identifier for Run-in and Run-out	N
RUN_IN_OUT_TYPE	Short, 1	There are three types of features a) 1 = Run-in b) 2 = Run-out c) 3 = Run-in & Run-out	N
PARKING_ID	String, 255	ID of off-street public car park	Y
BUILDING_ENAME	String, 255	English names of Building of off-street parking site	Y
BUILDING_CNAME	String, 255	Chinese names of Building of off-street parking site	Y
CRE_DATE	Date	To store the date that the record was created.	N
RD_ID_1	Long, 6	The identifier of 1st intersected Road Centreline	Y
RD_ID_2	Long, 6	The identifier of 2nd intersected Road Centreline	Y
RD_ID_3	Long, 6	The identifier of 3rd intersected Road Centreline	Y
RD_ID_4	Long, 6	The identifier of 4th intersected Road Centreline	Y
REMARKS	String, 255	This field stores additional information.	Y
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N
BUILDING_ADDRESS	String, 255	Building of off-street parking site	Y

3.3.2.5. ROUNDABOUT

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
CRE_DATE	Date	To store the date that the record was created.	N
R_ABOUT_ID	Long, 6	The unique identifier of Roundabout. It can be used to link with external tables being maintained by other systems.	N
R_ABOUT_ENAME	String, 255	English names of Roundabout	Y
R_ABOUT_CNAME	String, 255	Chinese names of Roundabout	Y
R_ABOUT_TYPE	Short, 1	There are five types of Roundabouts: 1 = Typical roundabouts (default); 2 = Mini roundabouts; (The radius of the center part of the roundabout is less than two meters) 3 = Double roundabouts; 4 = Others; 5 = Spiral roundabouts;	N
GRADED	String, 1	Whether the Roundabout is grade separated	N
SIGNALIZED	String, 1	Whether the Roundabout is a signalised roundabout	N
NO_OF_ARM	Short, 1	Number of arms	N
RD_ID_1	Long, 6	The identifier of 1st Road Centreline	Y
RD_ID_2	Long, 6	The identifier of 2nd Road Centreline	Y
RD_ID_3	Long, 6	The identifier of 3rd Road Centreline	Y
RD_ID_4	Long, 6	The identifier of 4th Road Centreline	Y
RD_ID_5	Long, 6	The identifier of 5th Road Centreline	Y
RD_ID_6	Long, 6	The identifier of 6th Road Centreline	Y
RD_ID_7	Long, 6	The identifier of 7th Road Centreline	Y
RD_ID_8	Long, 6	The identifier of 8th Road Centreline	Y
RD_ID_9	Long, 6	The identifier of 9th Road Centreline	Y
RD_ID_10	Long, 6	The identifier of 10th Road Centreline	Y
REMARKS	String, 255	This field stores additional information.	Y
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N

3.3.2.6. PROHIBITION

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
PROHIBITION_ID	Long, 5	A unique identifier of storing the Prohibition feature	N
EXC_VEH_TYPE	String, 100	This field is used to indicate one or more vehicle types excluded by the Prohibition. Possible vehicle types include: a) ALL = All Motor Vehicle b) FB = Franchised Bus c) GMB = Green Mini Bus d) GV = Goods Vehicle e) NA = NA (Default) f) NFB = Non-franchised Bus g) OTH = Others h) PC = Private Car i) PLB = Public Light Bus j) RLB = Private Light Bus k) TX = Taxi	Y
INC_VEH_TYPE	String, 100	This field is used to indicate one or more vehicle types included by the Prohibition. Possible vehicle types include: a) ALL = All Motor Vehicle b) FB = Franchised Bus c) GMB = Green Mini Bus d) GV = Goods Vehicle e) NA = NA (Default) f) NFB = Non-franchised Bus g) OTH = Others h) PC = Private Car i) PLB = Public Light Bus j) RLB = Private Light Bus k) TX = Taxi	N
PART_TIME_PROHIBITION	String, 1	Part time Prohibition. Y for Yes; N for No.	Y
EFF_ALL_DAYS	String, 1	Effective for all days. Y for Yes; N for No.	Y
OTHER_REST_TYPE_GV	String, 100	Other type of restriction. The values include: a) NA b) Length c) Height d) Weight e) Axle Weight	Y

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
REMARKS	String, 255	This field stores additional information on Prohibition. This field can be used for displaying labels.	Y
ROAD_ROUTE_ID	Long, 6	This is the field that stores the Road Route identifier of the Road Route (ROUTE_ID) being referenced.	N
CRE_DATE	Date	To store the date that the record was created.	N
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N
BOUND	Long, 1	This field is used to indicate the location of the feature along the centerline. If it is located on the left hand side of the centerline with reference to digitized direction, the value is 1. If it is located on the right hand side of the centerline with reference to digitized direction, the value is -1. If it is located on both sides of the centerline, with reference to digitized direction, the value is 0.	N

3.3.2.7. PERMIT

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
PERMIT_ID	Long, 6	An unique identifier of storing the Permit feature	N
ROAD_ROUTE_ID	Long, 6	This is the field that stores the Road Route identifier of the Road Route (ROUTE_ID) being referenced.	N
REMARKS	String, 255	This field stores additional information on Permit. This field can be used for displaying labels.	Y
CRE_DATE	Date	To store the date that the record was created.	N
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N
BOUND	Long, 1	This field is used to indicate the location of the feature along the centerline.	N

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
		If it is located on the left hand side of the centerline with reference to digitized direction, the value is 1. If it is located on the right hand side of the centerline with reference to digitized direction, the value is -1. If it is located on both sides of the centerline, with reference to digitized direction, the value is 0.	

3.3.2.8. PEDESTRIAN_ZONE

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
ROAD_ROUTE_ID	Long, 6	This is the field that stores the Road Route identifier of the Road Route (ROUTE_ID) being referenced.	N
TIME_ZONE	String, 255	This field stores the time zone for the bus use only. It stores the abbreviation of Traffic Sign (E.g. 24, 7-19, 7:30-9:30).	N
EFFECTIVE_DAY	String, 8	This field stores information about the frequency of the road pass occurrence by the flag at each digit position. There are 8 digits in this field. E.g. YNNNNNNN. The 1st digit represents Monday, 2nd digit represents Tuesday and so on. The 8th digit represents Public Holidays which has the precedence over the others. 'Y' means the road pass occurs on that day of every week; 'N' means the road pass does not occur on that day of every week.	N
REMARKS	String, 255	This field stores additional information on Pedestrian Zone. This field can be used for displaying labels.	Y
PED_ZONE_ID	Long, 6	A unique identifier of storing the Pedestrian Zone feature	N
CRE_DATE	Date	To store the date that the record was created.	N
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N

3.3.2.9. NSR

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
VEHICLE_TYPE	Long, 1	This is the field where the type of vehicle is stored. Vehicle restriction includes: 1 – All motor vehicles 2 – Taxis 3 – PLBs 4 – Goods Vehicles 5 – Others	N
TIME_ZONE	Long, 1	This field stores information with regard to the time zone of the restriction. 1 – 24 Hours 2 – 8 am to 10 am and 5 pm to 7 pm 3 – 7 am to 7 pm 4 – 7 am to midnight 5 – Others	N
EFFECTIVE_DAY	Long, 1	This field stores information with regard to the effective day of the restriction. 1 – All days 2 – All days except Sundays and Public Holidays 3 – General Holidays (Sundays and Public Holidays) 4 – Others	N
CRE_DATE	Date	To store the date that the record was created.	N
NSR_ID	Long, 6	A unique identifier storing the Stop Restriction	N
ST_CODE_1	Long, 6	The street code of 1st Road Centreline	Y
ST_CODE_2	Long, 6	The street code of 2nd Road Centreline	Y
ST_CODE_3	Long, 6	The street code of 3rd Road Centreline	Y
ST_CODE_4	Long, 6	The street code of 4th Road Centreline	Y
ST_CODE_5	Long, 6	The street code of 5th Road Centreline	Y
ST_CODE_6	Long, 6	The street code of 6th Road Centreline	Y
REMARKS	String, 255	This field stores additional information.	Y
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N

3.3.2.10. BUS_ONLY_LANE

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
ROAD_ROUTE_ID	Long, 6	This is the field that stores the Road Route identifier of the Road Route (ROUTE_ID) being referenced.	N
TIME_ZONE	String, 255	This field stores the time zone for the bus use only. It store the abbreviation of Traffic Sign (E.g. 24, 7-19, 7:30-9:30)	Y
EFFECTIVE_DAY	String, 8	This field stores information about the frequency of the road pass occurrence by the flag at each digit position. There are 8 digits in this field. E.g. YNNNNNNN. The 1st digit represents Monday, 2nd digit represents Tuesday and so on. The 8th digit represents Public Holidays which has the precedence over the others. 'Y' means the road pass occurs on that day of every week, 'N' means the road pass does not occur on that day of every week.	Y
REMARKS	String, 255	This field stores additional information on Bus Only Lane. This field can be used for displaying labels.	Y
BUS_ONLY_LANE_ID	Long, 6	A unique identifier of storing the Bus Only Lane feature	N
CRE_DATE	Date	To store the date that the record was created.	N
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N
BOUND	Long, 1	This field is used to indicate the location of the feature along the centerline. If it is located on the left hand side of the centerline with reference to digitized direction, the value is 1. If it is located on the right hand side of the centerline with reference to digitized direction, the value is -1. If it is located on both sides of the centerline, with reference to digitized direction, the value is 0.	N

3.3.2.11. CENTERLINE

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
STREET_ENAME	String, 255	Street / Bridge / Tunnel Name in English	Y
STREET_CNAME	String, 255	Street / Bridge / Tunnel Name in Chinese	Y
ELEVATION	Short, 1	The relative elevation level of a road element when two or more roads overlap. Numbers are used to represent the following elevations: a) Zero (0) is the ground level (default); b) A negative number (i.e. –1, –2, –3…) represents underground level; c) A positive number (i.e. 1, 2, 3…) represents flyover level.	N
ST_CODE	Long, 5	This is the 5 digit street code from Lands Department (“LandsD”) named roads according to Road Centerline Common Spatial Unit. a) 10001-29999 = Streets with gazetted name b) 30001-39999 = Streets with ungazetted name c) 40001-59999 = Streets without name	Y
EXIT_NUM	String, 50	This field stores the Exit number of Strategic Routes as well as its associated Strategic Route Number. (e.g. 9A, 4 indicates Exit number 9A of Route No. 4)	Y
ROUTE_NUM	Short, 2	This is the field that store Strategic Route number of the road.	Y
REMARKS	String, 200	Additional information	Y
ROUTE_ID	Long, 6	Unique Identifier for the Route segment	N
TRAVEL_DIRECTION	Short, 1	The permitted direction(s) of travel. Numbers are used to represent the following travel directions: 1 = Travel is permitted in both directions (default); 3 = Travel is only permitted in the digitised direction of the line feature	N
CRE_DATE	Date	To store the date that the record was created.	N
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N
ALIAS_ENAME	String, 255	Alias name for Street / Bridge / Tunnel in English	Y
ALIAS_CNAME	String, 255	Alias name for Street / Bridge / Tunnel in Chinese	Y

3.3.2.12. TURN

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
EDGE1END	String, 1	Indicates if the turn passes through the end of the first edge (Y means the turn passes through the end of the first edge, while N means the turn passes through the beginning of the first edge).	N
EDGE1FCID	Long, 2	The feature class identifier of the line feature representing the first edge of the turn.	N
EDGE1FID	Long, 6	The feature identifier of the line feature representing the first edge of the turn.	N
EDGE1POS	Double (8, 7)	The position along the line feature that represents the first edge of the turn. For a line feature that represents multiple edges, which can be created by lines with any vertex connectivity or points with override policy, the position indicates which of the feature's edge elements is the first edge in the turn.	N
EDGE2FCID	Long, 2	The feature class identifier of the line feature representing the 2nd edge of the turn.	N
EDGE2FID	Long, 6	The feature identifier of the line feature representing the 2nd edge of the turn.	N
EDGE2POS	Double (8, 7)	The position along the line feature that represents the 2nd edge of the turn.	N
EDGE3FCID	Long, 2	The feature class identifier of the line feature representing the 3rd edge in a multiedge turn with three or more edges.	Y
EDGE3FID	Long, 6	The feature identifier of the line feature representing the 3rd edge in a multiedge turn with three or more edges.	Y
EDGE3POS	Double (8, 7)	The position along the line feature that represents the 3rd edge of the turn with three or more edges.	Y
EDGE4FCID	Long, 2	The feature class identifier of the line feature representing the 4th edge in a multiedge turn with three or more edges.	Y
EDGE4FID	Long, 6	The feature identifier of the line feature representing the 4th edge in a multiedge turn with three or more edges.	Y
EDGE4POS	Double (8, 7)	The position along the line feature that represents the 4th edge of the turn with three or more edges.	Y
EDGE5FCID	Long, 2	The feature class identifier of the line feature representing the 5th edge in a multiedge turn with three or more edges.	Y

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
EDGE5FID	Long, 6	The feature identifier of the line feature representing the 5th edge in a multiedge turn with three or more edges.	Y
EDGE5POS	Double (8, 7)	The position along the line feature that represents the 5th edge of the turn with three or more edges.	Y
EDGE6FCID	Long, 2	The feature class identifier of the line feature representing the 6th edge in a multiedge turn with three or more edges.	Y
EDGE6FID	Long, 6	The feature identifier of the line feature representing the 6th edge in a multiedge turn with three or more edges.	Y
EDGE6POS	Double (8, 7)	The position along the line feature that represents the 6th edge of the turn with three or more edges.	Y
EDGE7FCID	Long, 2	The feature class identifier of the line feature representing the 7th edge in a multiedge turn with three or more edges.	Y
EDGE7FID	Long, 6	The feature identifier of the line feature representing the 7th edge in a multiedge turn with three or more edges.	Y
EDGE7POS	Double (8, 7)	The position along the line feature that represents the 7th edge of the turn with three or more edges.	Y
EDGE8FCID	Long, 2	The feature class identifier of the line feature representing the 8th edge in a multiedge turn with three or more edges.	Y
EDGE8FID	Long, 6	The feature identifier of the line feature representing the 8th edge in a multiedge turn with three or more edges.	Y
EDGE8POS	Double (8, 7)	The position along the line feature that represents the 8th edge of the turn with three or more edges.	Y
NO_TURN	Short, 1	Turn Restriction (Negative impedance implied that the turn is restricted. By default, -1 would be assigned.)	Y
CRE_DATE	Date	To store the date that the record was created.	N
REMARKS	String, 255	Additional information	Y
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N
TURN_ID	Long, 6	Unique identifier of Turn	N
EXC_VEH_TYPE	String, 100	This field is used to indicate one or more vehicle types excluded by the turn restriction. Possible vehicle types include: a) ALL = All Motor Vehicle b) FB = Franchised Bus c) GMB = Green Mini Bus d) GV = Goods Vehicle	N

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
		e) NA = NA (Default) f) NFB = Non-franchised Bus g) OTH = Others h) PC = Private Car i) PLB = Public Light Bus j) RLB = Private Light Bus k) TX = Taxi	
INC_VEH_TYPE	String, 100	This field is used to indicate one or more vehicle types included by the turn restriction. Possible vehicle types include: a) ALL = All Motor Vehicle b) FB = Franchised Bus c) GMB = Green Mini Bus d) GV = Goods Vehicle e) NA = NA (Default) f) NFB = Non-franchised Bus g) OTH = Others h) PC = Private Car i) PLB = Public Light Bus j) RLB = Private Light Bus k)TX = Taxi	N
PART_TIME_REST	String, 1	Part time restriction. Y for Yes; N for No.	N
EFF_ALL_DAYS	String, 1	Effective for all days. Y for Yes; N for No.	N
OTHER_REST_TYPE	String, 100	Other type of restriction. The values include: a) NA b) Length c) Height d) Weight e) Axle Weight	N

3.3.2.13. INTERSECTION

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
CRE_DATE	Date	To store the date that the record was created	N
INT_ID	String, 50	Unique Identifier for Intersection.	N
INT_TYPE	Short, 1	There are two types of Intersection Features a) 0 = Others (default) b) 1 = Signalized Junction	N
INT_ENAME	String, 255	English names of streets in the intersection or English name of the street where the point feature locate	Y
INT_CNAME	String, 255	Chinese names of streets in the intersection or Chinese name of the street where the point feature locate	Y
RD_ID_1	Long, 6	The identifier of 1st intersected Road Centreline	Y
RD_ID_2	Long, 6	The identifier of 2nd intersected Road Centreline	Y
RD_ID_3	Long, 6	The identifier of 3rd intersected Road Centreline	Y
RD_ID_4	Long, 6	The identifier of 4th intersected Road Centreline	Y
RD_ID_5	Long, 6	The identifier of 5th intersected Road Centreline	Y
RD_ID_6	Long, 6	The identifier of 6th intersected Road Centreline	Y
RD_ID_7	Long, 6	The identifier of 7th intersected Road Centreline	Y
RD_ID_8	Long, 6	The identifier of 8th intersected Road Centreline	Y
RD_ID_9	Long, 6	The identifier of 9th intersected Road Centreline	Y
RD_ID_10	Long, 6	The identifier of 10th intersected Road Centreline	Y
REMARKS	String, 255	Additional information.	Y
LAST_UPD_DATE_V	Date	To store the date that the record was updated for VASP.	N

3.3.2.14. ONSTREETPARK

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
FEATUREID	Long, 6	The unique identifier of On-street Parking	N
CRE_DATE	Date	To store the date that the record was created.	N
LAST_UPD_DATE	Date	To store the date that the record was updated.	N

3.3.2.15. TUN_BRIDGE_TOLL

Entity type: Data Table

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
TUNNEL_BRIDGE_NAME	String, 100	English name of Tunnel / Bridge	N
TUNNEL_BRIDGE_ CHINESE_NAME	String, 100	Chinese name of Tunnel / Bridge	Y
FEATURE_ID_1	Long, 6	Feature ID #1 of Tunnel / Bridge	Y
FEATURE_ID_2	Long, 6	Feature ID #2 of Tunnel / Bridge	Y
EFFECTIVE_DATE	Date	Effective date of toll rates	Y
GAZETTED_TOLL	Long, 3	Gazetted toll rates of Tunnel / Bridge	N
CONCESSION_TOLL	Long, 3	Concession toll rates of Tunnel / Bridge	N
VEHICLE_CLASS_ DESCRIPTION	String, 5	Description of vehicle type for the toll rate	N
REMARKS	String, 1000	Remarks of Tunnel / Bridge	Y
LAST_UPDATED_DATE	Date	To store the date that the record was updated.	Y

3.3.2.16. GISP_ON_STREET_PARKING

Entity type: Data Table

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
FEATUREID	OID	The unique identifier of On-street Parking	N
STREET_NAME_ENG	String, 100	English street name of On-street Parking	Y
STREET_NAME_CHI	String, 100	Chinese street name of On-street Parking	Y
METER_NONMETER	String, 1	To indicate whether meter is provided	Y
CAPACITY	Short, 1	Capacity of On-street Parking	Y
VEHICLE_TYPE_ DESCRIPTION_1	String, 100	Vehicle type (First) of On-street Parking	N
VEHICLE_TYPE_ DESCRIPTION_2	String, 100	Vehicle type (Second) of On-street Parking (if any)	Y
PARK_METER_PERIOD_ DESC	String, 100	Park meter period of On-street Parking	Y
OPERATIVE_HOUR_ENG	String, 600	English operative hour description of On-street Parking	Y
OPERATIVE_HOUR_CHI	String, 600	Chinese operative hour description of On-street Parking	Y
IRNP_REMARKS_ENG	String, 300	English remarks of On-street Parking	Y
IRNP_REMARKS_CHI	String, 300	Chinese remarks of On-street Parking	Y
X_COOR	Long, 6	X-coordinate of On-street Parking	N
Y_COOR	Long, 6	Y-coordinate of On-street Parking	N
LAST_UPDATE_DATE	Date	To store the date that the record was updated	N

3.3.2.17. TUN_BRIDGE_TV_TOLL

Entity type: Data Table

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
OBJECTID	OID	Object ID	N
TUNNEL_BRIDGE_NAME	String, 100	English name of Tunnel / Bridge	Y
TUNNEL_BRIDGE_ CHINESE_NAME	String, 100	Chinese name of Tunnel / Bridge	Y
FEATURE_ID_1	Long, 6	Feature ID #1 of Tunnel / Bridge	Y
FEATURE_ID_2	Long, 6	Feature ID #2 of Tunnel / Bridge	Y
EFFECTIVE_DATE	Date	Effective date of Toll Rates	N
DAY_OF_WEEK	String, 5	Day of week for Tunnel rates A = From Monday to Saturday (Excluding Public Holiday); B = Sunday and Public Holiday	N
START_TIME	String, 8	Starting time	N
END_TIME	String, 8	End time	N
GAZETTED_TOLL	Double, (3, 1)	Gazetted toll rates of Tunnel / Bridge	N
VEHICLE_CLASS_ DESCRIPTION	String, 5	Description of vehicle type for the toll rate	N
REMARKS	String, 1000	Remarks of Tunnel / Bridge	Y
LAST_UPDATED_DATE	Date	To store the date that the record was updated	Y

3.4. Unique Identifiers

The following table describes the unique identifier of each entity in the Transportation FSDT.

Data Layer	Unique Identifier	Description
RoadCentreLine	STREET CENTRELINEID	A unique identifier maintained by LandsD.
VEHICLE_ RESTRICTION	VR_ID	A unique vehicle restriction identifier maintained by TD.
TRAFFIC_FEATURES	FEATURE_ID	A unique traffic features identifier maintained by TD.
SPEED_LIMIT	SPEED_LIMIT_ID	A unique speed limit identifier maintained by TD.
RUN_IN_OUT	RUN_IN_OUT_ID	A unique run-in and run-out identifier maintained by TD.
ROUNDABOUT	R_ABOUT_ID	A unique roundabout identifier maintained by TD.
PROHIBITION	PROHIBITION_ID	A unique prohibition identifier maintained by TD.
PERMIT	PERMIT_ID	A unique permit identifier maintained by TD.
PEDESTRIAN_ZONE	PED_ZONE_ID	A unique pedestrian zone identifier maintained by TD.
NSR	NSR_ID	A unique no stopping restriction identifier maintained by TD.
BUS_ONLY_LANE	BUS_ONLY_LANE_ ID	A unique bus only lane identifier maintained by TD.
CENTRELINE	ROUTE_ID	A unique centreline identifier maintained by TD.
TURN	TURN_ID	A unique Turn identifier maintained by TD.
INTERSECTION	INT_ID	A unique intersection identifier maintained by TD.
ONSTREETPARK	FEATUREID	A unique on-street parking identifier maintained by TD.
TUN_BRIDGE_TOLL	OBJECTID	A system generated unique identifier maintained by TD.
GISP_ON_STREET_ PARKING	FEATUREID	A system generated unique identifier maintained by TD.
TUN_BRIDGE_TV_ TOLL	OBJECTID	A system generated unique identifier maintained by TD

3.5. Geographical Coverage

The dataset(s) specified in this document covers the entire territories of HKSAR.

3.6. Units of Measurement

The units of measurement used in all datasets under Transportation FSDT refer to the International System of Units ("SI").

3.7. Spatial Resolution/Scale

The spatial resolution(s) or scale(s) of the dataset(s) specified in this document is/are:

- Road Centreline: The dataset is maintained at a scale of 1:1,000.
- Road Network: The dataset is maintained at a scale of 1:1,000.

3.8. Spatial Representation Type

The spatial representation type(s) of the dataset(s) specified in this document is/are:

- All datasets are represented as points and lines to represent the geographical location of the entity.

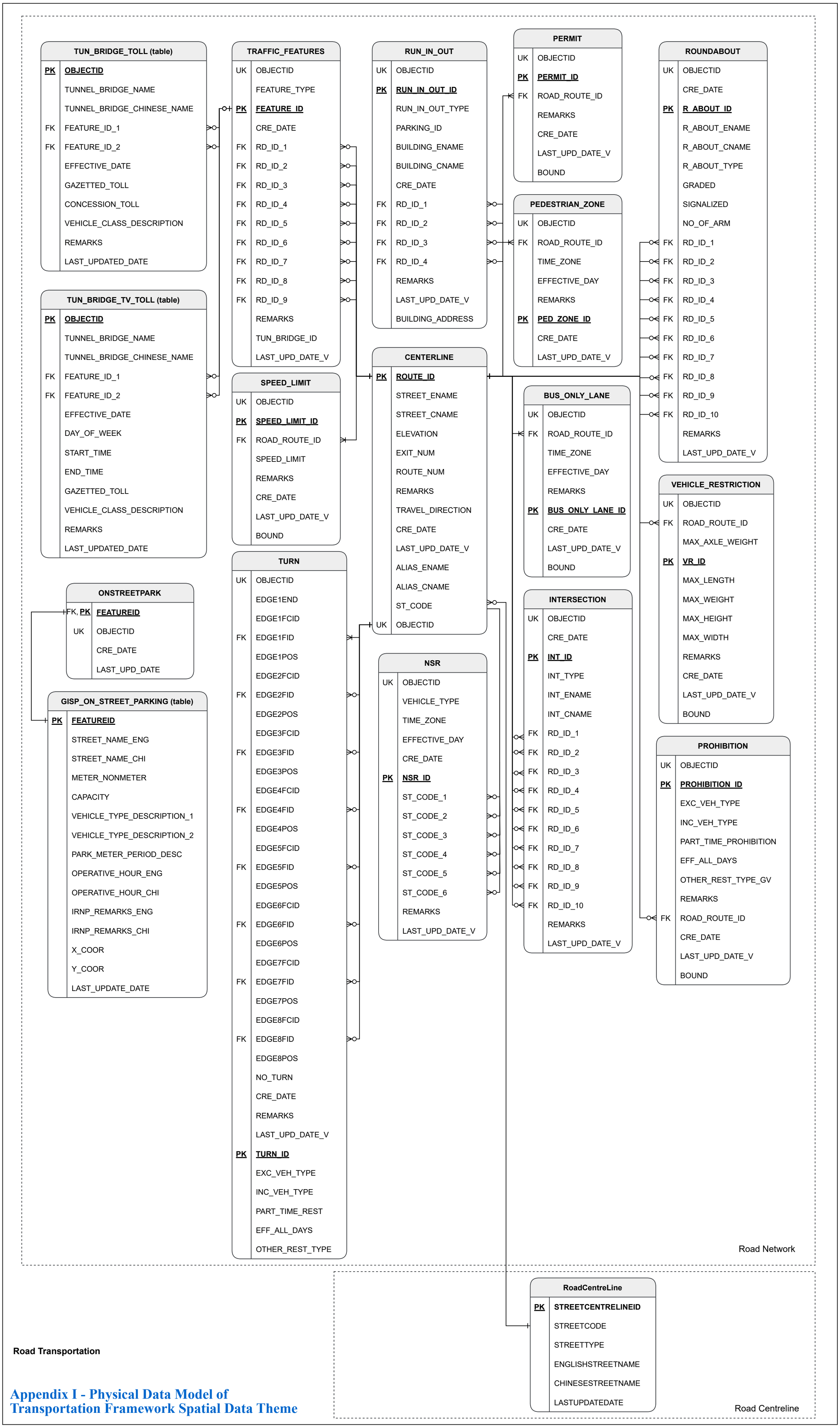
3.9. Spatial Reference System

The spatial data covered in this document are geo-referenced or projected to one of the Coordinate Reference Systems as specified in the Simplified Data Specifications for Coordinate Reference System of the CSDI.

3.10. Relationships and Dependencies with Other FSDTs

There are no relationships between Transportation FSDT and other FSDTs.

The English version shall prevail in case of any discrepancy or inconsistency between the English version and its Chinese translation.



Appendix I - Physical Data Model of Transportation Framework Spatial Data Theme